

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) What are the stages of site investigation? State objectives of each stage?		06
B) Describe, with a neat sketch, the wash boring method. In what field conditions would you recommend this method?		06
C) C.1) A SPT was conducted in a dense sand deposit at a depth of 22 m, and a value of 48 was observed for N. The density of the sand was 15 kN/m ² . What is the value of N, corrected for overburden pressure? C.2) Compute the area ratio of a thin-walled tube samples having an external diameter of 6 cm and a wall thickness of 2.25 mm. Do you recommend the sampler for obtaining undisturbed soil samples? Why?		06
Q.2 Solve <u>any two</u> following questions.		12
A) What are the common modes of bearing capacity failure of a footing. Explain in detail with Sketches?		06
B) A strip footing of width 3 m is founded at a depth of 2 m below the ground surface in a (c - ϕ) soil having a cohesion $c = 30$ kN/m ² and angle of shearing resistance $\phi = 35^\circ$. The water table is at a depth of 5 m below ground level. The moist weight of soil above the water table is 17.25 kN/m ³ . Determine (a) Ultimate bearing capacity of the soil, (b) Net bearing capacity, and (c) Net allowable bearing pressure and the load/m for a factor of safety of 3. Use the general shear failure theory of Terzaghi. $\phi = 35^\circ$ $N_c = 57.8$, $N_q = 41.4$ and $N_\gamma = 42.4$		06
C) If the water table in Q.2 B) rises to the ground level, determine the net safe bearing pressure of the footing. All the other data given in Q.2 B) remains the same. Assume the saturated unit weight of the soil γ_{sat} is 18.5 kN/m ³ .		06

Q. 3 Solve <u>any two</u> of the following.	12
A) What is swelling soils? Explain in detail the characteristics of swelling soils?	06
B) Define and explain the following terms: A) Swelling Potential B) Swelling Pressure C) Free Swell	06
C) Enlist & explain the essential steps involved in the final choice of the type of foundation?	06
Q.4 Solve <u>any two</u> of the following.	12
A) Classify and explain the types of Piles: a) Based on purpose b) Based on load transfer c) Based on method of construction	06
B) Enlist the different methods to estimate the load carrying capacity of piles and explain in detail static formulae method.	06
C) C.1) A timber pile was driven by a drop hammer weighing 30 kN with a free fall of 1.2 m. The average penetration of the last few blows was 5 mm. What is the capacity of the pile according to Engineering News Formula? C.2) A pile is driven with a single acting steam hammer of weight 15 kN with a free fall of 900 mm. The final set, the average of the last three blows, is 27.5 mm. Find the safe load using the Engineering News Formula?	06
Q. 5 Solve <u>any two</u> of the following.	12
A) Enlist the different methods for the analysis of finite slopes and explain in detail the stability number method.	06
B) Enlist and explain in detail the different types of slope failures?	06
C) An embankment is inclined at an angle of 35° and its height is 15 m. The angle of shearing resistance is 15° and the cohesion intercept is 200 kN/m^2 . The unit weight of soil is 18.0 kN/m^3 . If Taylor's stability number is 0.06, find the factor of safety with respect to cohesion.	06

*** End ***